Nile Odhiambo Okomo

Backend & Application Security Engineer

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SUMMARY

Backend engineer building payment, network and device management systems in Python and Go, with a focus on application security: secure code review, penetration testing and vulnerability remediation. Found and fixed a webhook authentication bypass, payout race conditions, SQL injection and privilege escalation in production systems. Founder of NetBill (115 registered ISPs; stable at a few hours a week). Seeking a full-time remote application security role.

EXPERIENCE

Founder & Lead Engineer, NetBill | Feb 2025 – Present | Remote
ISP billing and MikroTik provisioning platform | *Python, Flask, MySQL, Redis, Celery, FreeRADIUS*

- Built and run a platform for small ISPs to onboard routers, sell plans and collect M-Pesa and Paystack payments; 115 businesses registered.
- Designed CGNAT-friendly onboarding: routers register, pull configuration and send heartbeats over outbound connections only.
- Secure code review of NetBill's payment flows: fixed a webhook authentication bypass (spoofable CF-Connecting-IP), a duplicate payout race condition and a client-controlled payout amount.
- Encrypted router credentials and RADIUS secrets at rest; 650+ automated tests, one regression test per security fix.

Backend Engineer (Freelance), StrataxIQ | Mar 2026 – Present | Remote

- Quik-MDM (Go): wrote the Apple MDM protocol server for a multi-tenant Android and iOS device management platform, with a hash-chained, tamper-evident audit log verified hourly.
- TECCC (Python, Frappe): CRM backend for a German telecom reseller; added fail-closed HMAC webhook verification, closed a SQL injection vector and removed access control (RBAC) bypass switches.
- Footfall (Flask, FreeRADIUS): guest WiFi backend; fixed open redirect, SSTI and SSRF vulnerabilities and enforced RADIUS Message-Authenticator (Blast-RADIUS mitigation).
- b2bstore (ERPNext): replaced Administrator-level execution with a least privilege service account, verified by 257 per-role API access checks.

Backend Engineer (Freelance), Zuka | Aug 2025 – Oct 2026 | Remote
Event ticketing with instalment payments | *Python, Flask, MySQL, Redis, Celery*

- Built M-Pesa STK Push collections and B2C payouts, with KYC checks before organisers can withdraw.
- Prevented double booking with Redis locks and ticket forgery with HMAC-signed QR codes.
- JWT with refresh tokens, Google OAuth, rate limiting and schema validation; 260 automated tests.

Freelance Developer, Tweeps | Jan 2024 – Feb 2025 | Remote

- Built the website, an ordering platform and a payment dashboard for a fast-food restaurant; React, Next.js, Flask.

Software Engineer Intern, Hopewell | Feb 2023 – Jul 2023 | Remote

- Built a therapy booking platform with Node.js and Express in a 15-person team; wrote the requirements specification.

QA / Technical Writer, GrowthTribe | Nov 2022 – Feb 2023 | Remote

- User acceptance and API testing with Postman; wrote test protocols, API docs and user manuals.

PROJECTS

- Coffee Beans (Django, PostgreSQL), in development: hotel system with RBAC, maker-checker approvals and an immutable audit trail.
- Sheaf (Flask, M-Pesa), in testing: social commerce for SMEs; hardened payment callback authentication and fixed a settlement bug.

SKILLS

Security: secure code review, penetration testing, vulnerability remediation, OWASP Top 10, API security, access control, least privilege, Burp Suite, Nmap, Metasploit, privilege escalation, authentication and session design, RBAC, webhook and payment security, CVSS, MITRE ATT&CK

Languages and frameworks: Python (Flask, FastAPI, Django), Go, JavaScript, TypeScript, Node.js, Express, Frappe, ERPNext

Data and infrastructure: MySQL, PostgreSQL, Redis, SQLite, Elasticsearch, Celery, Docker, Nginx, systemd, GitHub Actions, Linux, Bash

Networking and payments: FreeRADIUS, MikroTik RouterOS, captive portals, Cloudflare, M-Pesa Daraja (STK Push, B2C, C2B), Paystack

EDUCATION

BSc Electronics, University of Eastern Africa, Baraton | Expected 2027

WRITING AND COMMUNITY

- Medium, medium.com/@beingnile (Jan 2022 – Present): 97+ article penetration testing series on Hack The Box, privilege escalation, SQL injection and OWASP Top 10. Training on PortSwigger Web Security Academy and pwn.college.
- Community Manager, SpaceYaTech (Jun 2022 – Apr 2023): taught data structures and algorithms to 300+ developers.
- Volunteer, GDG Kisii (Oct 2021 – Oct 2025): helped run Hacktoberfest, DevFest Western Kenya and workshops.